

WORLD DEVELOPMENT REPORT 2025

STANDARDS FOR DEVELOPMENT

Concept Note

“无规矩不成方圆” • “Without standards, no boundaries are set.”

—From the works of Chinese Confucian philosopher Mencius and his students

“He who owns the *standards*, owns the market.”

—Werner von Siemens, German electrical engineer and entrepreneur

“... there is again an abundance of concurrent contemporary testimony about the influence of entrepreneurial attitudes and *technical standards* on business performance that one would be ill advised to dismiss except on the strongest grounds.”

—David Landes in *The Unbound Prometheus* (1969)

“*Standards* are essential for all human activity, but most people take them for granted. Only when products fail to work, or mishaps occur, does the average person think about standards. Even in business, where money is at stake, standards are often given a low priority. There is a clear need ... for greater attention to standards.”

—U.S. Congress report entitled *Global Standards: Building Blocks for the Future* (1992)

“Basically, problems of standardization are the same whether they arise in the context of developing or developed countries. It is their detailed handling that may differ in the context of the prevailing socio-economic conditions and political circumstances.”

—Lal C. Verman in *Standardization—A New Discipline* (1973)

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Abbreviations

ABNT	Brazilian National Standards Organization
ACCSQ	ASEAN Consultative Committee for Standards and Quality
AI	artificial intelligence
ANSI	American National Standards Institute
ARSO	African Regional Organization for Standardization
ASEAN	Association of Southeast Asian Nations
ASME	American Society of Mechanical Engineers
BCBS	Basel Committee on Banking Supervision
BIS	Bureau of Indian Standards
BIS	Bank for International Settlements
BRSS	Bank Regulation and Supervision Survey
BSTI	Bangladesh Standards, Testing Institution
CE	Conformité Européenne
CEN	European Committee for Standardization
CENELEC	European Committee for Electrotechnical Standardization
CHWs	Community Health Workers
COPANT	Pan-American Standards Commission
DIN	German Institute for Standardization
EASC	Euro-Asian Council for Standardization, Metrology, and Certification
EC	European Commission
ESG	environmental, social, and governance
ETSI	European Telecommunications Standards Institute
EU	European Union
EUDR	EU Deforestation Regulation
EV	electric vehicle
FAO	Food and Agriculture Organization (United Nations)
FDI	foreign direct investment
FRAND	fair, reasonable, and non-discriminatory
GDPR	General Data Protection Regulation (European Union)
GFSI	Global Food Safety Initiative
GHG	greenhouse gas
GPT	general purpose technology
HRM	human resources management
ICAO	International Civil Aviation Organization
ICTs	information and communication technologies
IEC	International Electrotechnical Commission
IEEE	Institute of Electrical and Electronic Engineers
ILO	International Labour Organization
IoT	Internet of Things
IP	intellectual property
IPPC	International Plant Protection Convention
IPRs	intellectual property rights
IPSAS	International Public Sector Accounting Standards
ISEAL	International Social and Environmental Accreditation and Labeling Alliance
ISO	International Organization for Standardization
ISPMs	International Phytosanitary Measures
IT	information technology
ITU	International Telecommunication Union

KYC	Know Your Customer
LIMS	Laboratory Information Management System
MEPS	minimum energy performance standards
MQS	minimum quality standard
NQI	national quality infrastructure
NSBs	National Standards Bodies
NTMs	non-tariff measures
OECD	Organization for Economic Co-operation and Development
OIML	International Organization for Legal Metrology
PFM	public financial management
PIRLS	Progress in International Reading Literacy Study
PISA	Programme for International Student Assessment
PPP	public-private partnership
QI	quality infrastructure
QR	quick response
SARSO	South Asian Regional Standards Organization
SDGs	Sustainable Development Goals
SDO	standards development organization
SEP	standard essential patent
SPS	sanitary and phytosanitary
SVR	stolen vehicle recovery
TBT	technical barriers to trade
TCP/IP	Transmission Control Protocol/Internet Protocol
TIMSS	Trends in International Mathematics and Science Study
TPCS	third-party certification systems
UPC	Uniform Plumbing Code
UPU	Universal Postal Union
W3C	World Wide Web Consortium
WDR	World Development Report
WHO	World Health Organization
WOAH	World Organisation for Animal Health
WTO	World Trade Organization

Part I: Why Standards for Development?

Why do standards matter, for what, and for whom? Who sets standards, and why does that matter? What roles do standards play in the progress of nations from low levels of development to the highest?

The *World Development Report 2025* (the “Report” or “WDR 2025”) will argue that setting and enforcing standards across the economy, society, environment, and government—and periodically revising and raising standards—are the heart of the development process, which is, in essence, all about improving living standards. Consequently, developing countries can make substantial gains by choosing and implementing the right standards effectively. Conversely, a poor choice of standards, their weak implementation, or exclusion from participation in the process of setting standards on the global stage can inflict large costs.

So how can developing countries leverage standards to foster sustainable growth? WDR 2025 will explore how standards can be used as a tool for development—to improve the productivity of firms, enable job creation, conduct mutually beneficial trade, enhance public access to basic social services, widen financial inclusion, or limit the adverse impacts of economic activities on the environment—and speed progress toward the Sustainable Development Goals. Standards can also improve the capacity of the public sector, for example, by enhancing the ability to recruit talented civil servants, ensure the most efficient use of public resources, and effectively enforce public policies.

Overview of the Report

Standards play a role that is simultaneously central and unsung in our modern world. From the first attempts to unify weights and measures in ancient empires to today's complex web of international protocols governing everything from the internet to food safety, standards have been the bedrock upon which societies have built their economic, political, and social structures.

WDR 2025 defines *standards* as codified solutions that are designed for common and repeated use and are recognized by multiple stakeholders (de Vries 1997). Railroad gauges, the kilogram, and building codes are all examples of standards. While most standards are voluntary, often developed by private organizations on behalf of their stakeholders, standards become mandatory when enshrined into law (as technical regulations).

In the process of economic development, productive sectors in countries transition from an *Absence* of standards, a situation without standards or a myriad of local standards, to a situation where they *Adopt* international standards, either by taking them as-is or adapting them to local economic contexts, capabilities, and ambition, and finally to one where they *Author* new international standards, often reflecting a country's economic goals and political priorities. “Authoring” standards in WDR 2025 refers to the participation of countries in the international standard-setting process. As countries and sectors navigate these transitions, they face several trade-offs, such as when to set standards, and how to define their scope and their stringency. Setting standards prematurely risks countries getting locked in with suboptimal results, while setting them too late may limit their effectiveness. Setting too wide a scope may lead to unnecessarily cumbersome requirements, while setting the scope too narrowly may limit the impact. Compliance costs increase with stringency, so setting the standard too high can lead to exclusion, while setting it too low can forsake the benefits of standardization. Based on their capabilities and resources, countries need to manage these trade-offs to achieve higher quality *and* aggregate welfare—the essence of development.

This Report is the first *World Development Report* devoted to standards, but it builds on the work of many previous WDRs (refer to appendix B). The Report will mainly focus on the transitions from *Absence* to *Adoption* and *Adaptation* because they are the most relevant for the largest share of developing economies. However, it will also address how developing countries can meaningfully participate in ongoing international standard-setting processes in emerging technologies (*Authoring*).

The Report will be organized into four parts. Part I will review the historical contribution of standards to development (both positive and negative) over the past century and a half and will illustrate how countries adopted standards. Examples will be drawn from the areas of metrology (the scientific study of weights and measures) and global trade; the rise of general purpose technologies (GPTs) like electricity and railways in the late nineteenth century driving the need for compatibility standards; the development of food safety standards, focusing on the Codex Alimentarius Austriacus within the Austro-Hungarian Empire—the forerunner of the Food and Agriculture Organization’s Codex Alimentarius (Food Code); the rise of the United States as an industrial powerhouse in the period between the first and second world wars by leveraging standards to achieve scale; the economic unification of India following independence; and the ascendance of China as a global manufacturing leader in the twenty-first century.

Part II will define standards and provide a taxonomy. WDR 2025 views standards not merely as the codification of technological knowledge and expertise but also a form of collective wisdom and global cooperation. This part of the Report will present the rationale for standards based on economic principles: to help correct market failures arising from imperfect information, coordination failures, and externalities. It will also provide the rationale for what standards the Report will cover, and those that it will not. It will examine when and how standards can be used to support regulation and when markets should be left alone. It will also analyze how developing countries can navigate the complex landscape of standards, choosing how to adopt them and when to use their countries’ limited resources to participate in the development of new standards. When should countries that lack standards adopt international standards as-is or adapt them to local conditions? How does this choice vary by different levels of development and across sectors, from medicine to machinery? When should standards be normative or advocative (voluntary) or prescriptive and enshrined in law (mandatory)?

Part II will also consider the key challenges that developing countries face when trying to have a “seat at the table” and influence the setting of international standards in areas where they have a particular interest or for emerging technologies—from artificial intelligence (AI) to successive generations of cellular network technology such as 5G/6G—that are increasingly important for development but are also at the heart of geopolitical competition. Almost without exception, emerging technologies have been developed in industrialized countries and, more recently, China. In choosing among competing standard regimes, developing countries face lock-in effects with long-term economic and political consequences. Some countries with increased technical capabilities and economic and political clout might author new standards and expect other countries to adopt them. This progression from adapting to adopting to authoring is not linear or universal. Countries may simultaneously adopt, adapt, and author standards in different sectors based on their specific needs and capabilities.

Part III will evaluate the potential gains and trade-offs of standards in select areas of development. The aim is not to be exhaustive or to provide in-depth analysis of a particular sector or aspect of development. Rather, WDR 2025 will discuss how many development problems can be addressed with standards—sometimes in combination with other policies—and highlight that the trade-offs policy makers face when setting and enforcing standards are common across different areas of development.

Part III will be organized along four main areas: (1) standards for the economy, including production, trade, investment, finance, and technology; (2) standards for the society, including in basic services such as education and health care, as well as labor standards and skills certification at both national and international levels; (3) standards for the environment, including clean water, air, energy, and waste management; and (4) standards for state capacity, including human resource management, public financial management, and procurement in the public sector. For each of these four areas, Part III will present empirical evidence of the impact of specific standards on development outcomes in that sector. It will address when standards are the appropriate instrument, including the scope and design of the instrument. It will also consider cross-cutting areas, including gender and data.

Part IV will discuss the policy implications for economies facing different economic circumstances and at varying stages of development. It will pay special attention to the imperatives for middle-income countries.

Standards *in* development

Imagine a world without standards. Your smartphone won't connect to any Wi-Fi network. Your car can't be easily repaired because it requires unique and irreplaceable parts. Buyers and sellers struggle to agree on the quality, quantity, or value of goods and services, leading to frequent disputes and mistrust. International trade is costly and sluggish because every product "speaks" a different technical language. Any economic activity results in high transaction costs of acquiring information, searching, contracting, bargaining, monitoring, enforcing, and managing disputes (Coase 1960; Commons 1931; Williamson 1979). Standards provide order, and a world without standards is a world of inefficiency—or chaos.

Standards are the invisible infrastructure that allows economies to communicate, innovate, and grow. While standards and innovation might sometimes appear to be opposites, they are invariably complements. Innovation is based on standards and drives the creation of new standards (Hawkins, Blind, and Page 2017). Standards are so fundamental that we rarely notice them, yet their absence would completely unravel the complex tapestry of modern life. The story of standards is the story of human progress itself—a testament to our remarkable ability to create order from chaos, bridge differences, and imagine collective solutions to collective challenges.

Standards play a crucial role in economic development by setting goals, establishing priorities, measuring performance, and driving change. They are essential for interoperability and technical harmonization, which lower coordination and transaction costs, mitigate risks, and enhance consumer confidence. By facilitating economies of scale in manufacturing and promoting compatibility between complementary products, standards create efficiencies both within and across firms and industries. They enable the harmonization of measurement and calibration, coordinate suppliers to adopt compatible solutions along the supply chain, signal product quality and performance, boost production and thus reduce unit costs, promote safety, and leverage network effects, particularly in information and communication technologies (ICTs).

Standards aren't just about commerce. They are the invisible threads that weave the fabric of our societies. When the French Revolution handed the task of standardization to its scientists, it was not merely a technical decision. It was a political statement, a declaration that rational, universal systems could form the basis of a new social order. This idea—that shared standards could unite a nation—has echoed through history.

The Industrial Revolution marked a turning point in the history of standards. As production scaled and trade expanded, the need for interchangeable parts and consistent quality became paramount. It was in this crucible that many modern standards bodies were born. The British Standards Institution emerged within the Institution of Civil Engineers in 1901 to tackle the chaos of incompatible industrial components. Its work laid the foundation for the international standards movement that would follow—a movement that has been instrumental in building the interconnected global economy we know today (refer to appendix C for a list of major standard-setting bodies).

In today's digital age, standards have taken on added importance. With the advent of the internet, the massive amount of data flowing across devices and geographies have led to a myriad of new standards that will only grow with the emergence of new technologies such as machine learning, artificial intelligence (AI), the Internet of Things (IoT), robotics, and so on. The internet, that great enabler of global communication and commerce, is itself a triumph of standardization. Every email sent and every webpage loaded relies on a complex ballet of agreed-upon protocols. These digital standards have not only facilitated unprecedented economic growth but have also reshaped our social and political landscapes, creating new forms of public discourse and political influence, and challenging traditional notions of governance. As we stand on the cusp of transformative technologies like artificial intelligence and quantum computing, the role of standards becomes even more crucial. They will determine not just how technologies interact but how societies evolve.

But setting standards is not without challenges. Their creation is costly. Developing a global standard for, say, 5G telecommunications requires massive investments. Companies and governments must collaborate to create something bigger than themselves, even as their instincts and incentives all push them toward secrecy and subterfuge. Standards also generate positive externalities—benefits that extend beyond the immediate users. When a country adopts international accounting standards, for example, it not only improves its financial reporting but also enhances global economic transparency and facilitates cross-border investments. Conversely, standards can also be a source of excessive market power, particularly when they are embedded in patents.

Standards also require an appropriate ecosystem that stimulates their compliance. What is the value of a standard if it is not being used or enforced? This is where state capacity and governance become critical and why government engagement and industry collaboration in developing and enforcing standards becomes necessary.

Part II: Conceptual Framework

Typology of standards

WDR 2025 focuses on three types of standards (David 1987). First, *compatibility standards* are agreed-upon specifications that ensure interoperability and compatibility between different systems, devices, or components. These standards enable products or services to work together seamlessly even if developed by different manufacturers. The railroad standard gauge and standardized freight containers are both examples of compatibility standards.

Second, *measurement standards* are agreed-upon methods to ensure consistency, accuracy, and reliability in measurement. Put simply, they define a “ruler,” epitomized by the field of metrology. Measurement standards play a crucial role in various fields, including health care, education, industry, and commerce, by facilitating interoperability, reducing errors, and enhancing consumer confidence in products and services.

They enable accurate comparisons of measurements across different locations and times, thus supporting global trade and technological advancement.

Third, *quality standards* define requirements for products, services, processes, or systems, such as management systems (de Vries 1998). Examples of quality standards include product safety standards, caps on fuel emissions, professional skill qualifications, and licensing and certification requirements.

Why adopt standards?

Standards can play a crucial role in addressing market failures and thereby achieve various developmental goals. As table 1 summarizes, standards are mainly used to address three main market failures: *imperfect information* (in contexts with incomplete or asymmetric information); *coordination failures* (when actors fail to coordinate even though it would benefit everyone); and *negative externalities* (when actors impose costs on others without bearing the consequences). The role of standards may be particularly important in developing economies because they face higher transaction costs (Dixit 2003) and greater market failures (Stiglitz 2010), and have weaker state capacity (Laffont 2005).

Table 1 How standards address market failures

	Role of standards	Outcomes	Impact (examples)
Imperfect information	Provide transparent, reliable, or consistent information	Lower search and transaction costs	- Lower uncertainty - Trust - Consumer protection - Efficiency
	Offer codified good practices	Transfer of knowledge	Effectiveness, productivity, quality upgrading
Coordination failures	Provide common specifications (e.g., for interfaces, definitions, measurements)	- Economies of scale - Compatibility - Lower transaction costs and barriers to entry - Fewer varieties	- Enable network industries - Increased market sizes - Improved usability of tech - Efficiency
Negative externalities	Define thresholds, processes, or technologies to reduce harm to third parties (e.g., society or the environment)	Common and verifiable way of including externalities	- Environmental protection - Social sustainability - Economic fairness

Source: WDR 2025 team.

Standards respond to imperfect information in two ways. First, they provide transparent, reliable, and consistent information, which can lower search and transaction costs. For example, buyers often lack the necessary information to judge the quality of a certain product such as food or medicines. Standards related to product quality can bridge the information gap. Standards can also be used for producers to signal that their products are of a certain quality. They can claim this themselves, or they may involve a certification body to have an independent assessment. For example, an ISO 9001 certificate signals that a company's quality management system meets the requirements specified in this standard, reducing the need for extensive and costly quality checks by customers (Clougherty and Grajek 2014; Manders 2015). Certification can provide access to high-value customers, but it can be costly for many firms in developing countries (Verhoogen 2023).

Second, many standards are the result of the pooling of expertise from leading experts agreeing on the best way of doing things, particularly in the case of international standards. A standard therefore offers codified good practices that have informational value to its users. Standards can play an important role for the transfer of knowledge and contribute to quality upgrading and increases in productivity. These are benefits that developing countries may be able to reap by using international standards.

Standards promote coordination by offering common specifications. Such standardization facilitates economies of scale, reduces transaction costs, and lowers barriers to entry, among other benefits. Standards help enable network industries (such as railways, telecommunications); increase market size (for example, via regional economic integration, as in the European Union, EU); and improve efficiency. They also increase the usability of technology, allowing users to switch between compatible products.

Following the rapid increase in the adoption of the metric system (as a measurement standard) in the mid-nineteenth century, global trade increased from about 2 percent of global GDP in the 1820s to 17.5 percent by 1870 and continued to grow steadily (Ortiz-Ospina, Beltekian, and Roser 2023; USMA 2024). More recently, the adoption of the metric system by several US companies in the early 1970s, including Caterpillar Tractor Company and General Motors, allowed better coordination with metric-based countries. Implementing a unified measurement system allowed these companies to streamline their production process and substantially reduce the variety of sized parts required, resulting in lower inventory costs and costs of serving international markets. The overall expenses associated with the metrication process were minimal compared to the substantial benefits realized (United States, NIST 2022).

Establishing standards can promote fair competition by lowering entry barriers, encouraging firms to offer competitive prices, and expanding access to their goods and services. For example, compatibility standards in telecommunications or payments allow users of one network to connect with users of another, promoting competition among service providers (Arabehty et al. 2016; Beck and De La Torre 2007; Crémer et al. 2023). Minimum quality standards also boost market competition. Conformité Européenne (CE) standards, for instance, enable smaller manufacturers to compete by ensuring they meet the same safety requirements as established brands (Takhar 2020). Similarly, bioequivalence standards allow generic drug manufacturers to compete with brand-name drugs, assuring consumers that generics are just as effective (ScienceDirect 2008; Smith Marsh 2023).

Standards are also useful to offset externalities. For example, pollution is a negative externality, where one firm's air pollution impairs the health of the surrounding population. If a government seeks to reduce carbon emissions, one way to do so is to impose a minimum standard for clean air. Standards can be used to define thresholds, processes, or technologies to reduce harm to third parties. For example, ISO 14064 provides standards for quantifying, monitoring, and reporting greenhouse gas emissions and removals, thereby helping entities mitigate their environmental impact (Wiegmann, de Vries, and Eom 2023). Governments can enact regulations to enforce such a standard, adopting the international benchmark from the International Organization for Standardization (ISO). In the realm of finance, lending standards and capital requirements set minimum conditions for financial institutions to extend credit, reducing the risk of financial instability from excessive lending.

The many benefits of standards notwithstanding, WDR 2025 will also highlight the pitfalls of standards. Dominant firms can push their own technological standards to entrench their market power. Higher standards can be a case of the best being the enemy of the good if they lead to higher costs of adoption and enforcement. Standards can lock firms and countries into a suboptimal equilibrium if they prevent them from adjusting to a rapidly changing environment, as was the case with an international monetary system based on the "Gold Standard." Standards can also discourage product differentiation, and in some cases can exacerbate externalities (for instance, if firms design products that are "just good enough" to meet an environmental standard) (Houde 2022). And there is considerable evidence that countries are using standards as non-tariff barriers to shut out imports from lower-income countries.

Scope of the Report

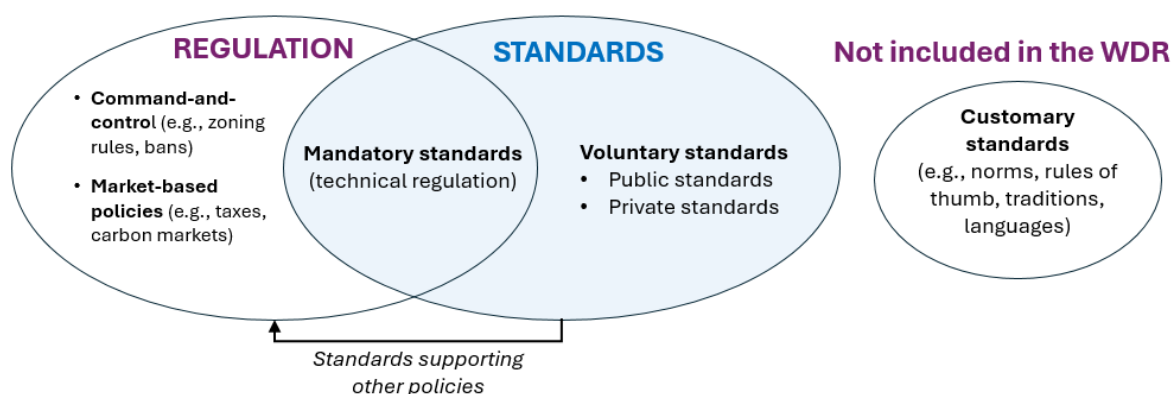
WDR 2025 will explore both voluntary and mandatory standards (as shown in the light blue oval in figure 1). Voluntary standards are typically developed by private standard development organizations at the initiative of stakeholders. Voluntary standards can also be developed by public organizations to achieve public policy goals but remain nonbinding. While standards are often considered voluntary in nature, the WDR 2025 notably emphasizes the role of mandatory standards for which compliance is required by law. Such mandatory standards are called technical regulations—legally binding documents specifying the characteristics of a product or its production methods (Kellermann 2019).

Governments can regulate markets through market-based policies (that is, price incentives) without directly controlling technical specifications, quantities, or allocations. Examples include carbon taxes, tradable emissions permits, housing or school vouchers, subsidies, and tax cuts associated with specific activities. Alternatively, governments can use “command and control” policies directly mandating what is legal and illegal. Examples are licensing and permits to operate as a business or service provider, bans, restrictions (such as those on food safety or labor conditions or pesticides), and price controls (such as minimum wages or usury laws).

While not all market-based or command-and-control regulations are standards themselves, standards are key to implementing a wide variety of policies. For example, market-based policies may be used to ease enforcement by providing fiscal incentives to help firms adopt green technologies and thus comply with environmental standards or, more generally, to reward firms that produce above the standard and penalize those below it (Fischer 2019). Governments can also play a role in ensuring that industry has an enabling environment to set standards.

Given the vast universe of standards, it is important to establish what the WDR 2025 will not cover. It will not address all types of standards; it will not include customary standards such as unwritten norms, conventions, rules of thumb, traditions, languages, or standards set by a private entity that apply only to itself.

Figure 1 Scope of the Report: Voluntary standards, mandatory standards, and regulation



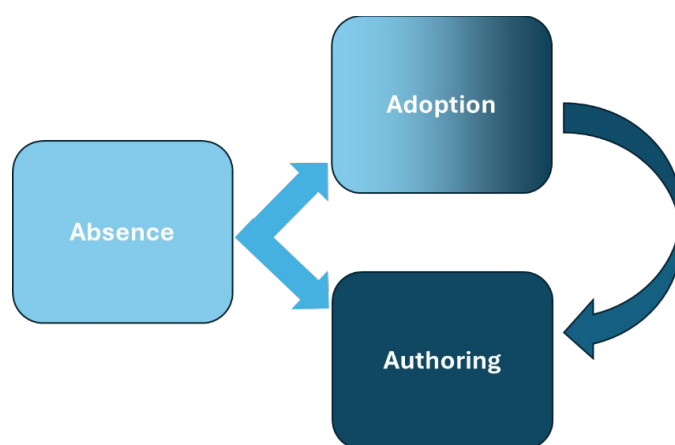
Source: WDR 2025 team.

Note: WDR = World Development Report.

Transitioning toward higher standards

WDR 2025 advances the idea that standards are central to economic development. Countries typically move from an *Absence* of standards to a state where domestic sectors *Adopt* a standard, either by accepting an international standard as-is or by aligning national standards to international standards by adapting international standards to local economic contexts and capabilities, or directly to a state where countries *Author* new international standards, often reflecting a country's political priorities and ambitions (refer to figure 2). “Authoring” standards in WDR 2025 refers to participation in the international standard-setting process. Countries can also transition from the *Adoption* of standards to *Authoring* standards. Table 2 illustrates the transitions from *Absence* to *Adoption* and from *Absence* or *Adoption* to *Authoring* with examples, highlighting the necessary conditions as well as the possible risks during these transitions.

Figure 2 Transitioning toward standards



Source: WDR 2025 team.

The journey of standards—from adoption to authoring—reflects a country's developmental trajectory, and it need not be linear. Countries can move between the choices of adoption, adaptation, or authorship in different sectors as their circumstances change, and may simultaneously engage in all three processes across different sectors based on their specific needs and capabilities. This journey embodies the complex interplay between scientific and technological advancements, social acceptance, economic pragmatism, business interests, and political ambition and power. Understanding this process is crucial for policy makers, businesses, and civil society as they navigate the increasingly interconnected and complex global landscape of standards.

Standards adoption is fundamentally a social, political, and commercial process. While it may reflect a country's readiness and willingness to align with international norms and commercial imperatives, it may also reflect a resignation to external pressures and lack of alternatives. Consider the case of the Philippines adopting International Public Sector Accounting Standards (IPSAS) (IFAC 2019). This was not just a technical decision; it was a social compact—a declaration of intent to speak the same financial language as the global community. The catalyst? A desire for improved transparency and international credibility.

Table 2 The Transition from Absence to Adoption and Adoption to Authoring: Examples, necessary conditions, and risks

Transitions	Examples	Necessary conditions	Risks
Absence to Adoption			
The adopted standard is the international standard.	<i>Alignment with international standards, especially in export-oriented sectors.</i>	- Functioning NQI system with critical mass of domestic producers that can conform to international standards.	- Potential exclusion of some domestic and informal producers, especially if there is no grace period for adjustment or the standard is too stringent for local conditions.
	Examples: EU accession; coffee sector Ethiopia.	- Expertise to make products and services compatible with international standards.	- Lack of local consultation during the adoption process may lead to misalignment with local conditions.
	<i>Adoption of international standards for emerging technologies and in network industries.</i>		- Lack of adequate quality infrastructure to assess conformity with international standard.
	Examples: Internet TCP/IP protocols, QR codes.		
	<i>Alignment with international measurement conventions.</i>		
	Examples: GDP.		
Absence to Adaptation			
The adopted standard is different from the international standard.	<i>Modification of standards to meet local conditions such as climate, voltage, etc. (one size does not fit all).</i>	- Technical capacity to modify international standards.	- Trap or lock-in. The less stringent standard adapted becomes the long-term standard without any positive impact if there is not a time- bounded transition to the global standard.
	<i>International standard is too stringent given consumer demand, productive capacity, and enforcement capacity.</i>	- Some quality and conformity infrastructure.	- Capture. Alterations reflect local industry interests but do not represent the common good.
	Examples: Environmental standards such as for air and water quality.		- Costs of conformity increase due to divergence from international approaches.
Absence or Adoption to Authoring			
A new standard is created.	<i>Any standard where the country is the leader or dominant producer.</i>	- Political ambition and clout to influence other countries to accept the standard authored.	- Market power.
	Examples: Emerging technologies (EVs, IoT, 5G); Teff grain quality standard (Ethiopia authored the international standard).	- Having sectors with technological leadership and capacity to participate in international standards development.	- Decoupling as leading countries impose their own standards on others.
		- Having concentrated production of a good or a technology.	- Authoring deviating national standards increases fragmentation as countries must choose which standards to accept, leading to slower technology diffusion and spheres of influence.
			- When moving from absence to authoring, actors may lack capacity to influence international standards, including references with other international standards.

Source: WDR 2025 team.

Note: EU = European Union; EVs = electric vehicles; IoT = Internet of Things; NQI = national quality infrastructure; QR = quick- response; TCP/IP = Transmission Control Protocol/Internet Protocol.

a. National quality infrastructure (NQI) refers to the institutional framework for standardization, conformity assessment, metrology, and accreditation.

Adoption is not always a voluntary process. History offers a stark reminder in the form of colonialism—perhaps the most extensive and often coercive transfer of standards in human history. From language to legal systems, colonial powers imposed their standards on vast swathes of the globe, reshaping societies in their image. This legacy continues to influence global standard adoption today, highlighting the complex interplay between power, culture, and standardization. But even when standards are adopted voluntarily, their success depends on the capabilities of the industry, sector, country, or other adopter of the standard, as they must have the technical expertise, infrastructure, and resources to implement and enforce standards effectively.

In the transition from absence to adoption, compatibility standards and some measurement standards are established through expertise and collaboration among stakeholders. Examples include internet protocols such as TCP/IP (Transmission Control Protocol/Internet Protocol) and QR (quick-response) codes, which have become integral to modern communication and commerce. This transition requires a foundational level of expertise to ensure product compatibility with established standards. However, adoption in principle is different from widespread adoption in practice. Even a QR code requires investing in scanners that small shopkeepers might not be able to afford. While this phase can lead to increased efficiency and interoperability, it also carries risks such as nonconformity, because producers or stakeholders lack the expertise to meet the standard, which can undermine the effectiveness of the newly adopted standards.

As for quality standards, the transition to adoption typically comes in response to crises or demand shocks, highlighting the need for standardization—but without the will or capacity to fully implement an internationally established standard. For instance, environmental standards related to air and water quality often emerge as societies recognize the urgent need to protect public health and the environment. Necessary conditions for this transition include a triggering event that creates a demand for standards and the realization that existing international standards may be too stringent given local consumer demand and production capacities. This phase has inherent risks, such as the possibility that the adapted, less stringent standard may become entrenched as the long-term norm or that local industries may capture the standard-setting process, leading to overly lenient regulations and undermining the value of the standard.

Does one size fit all?

Ideally, countries would greatly benefit from adopting the common international standard as-is because it is widely recognized and trusted. In practice, however, consumer demand, production capabilities, state capacity, and other factors may lead countries to adapt the more stringent international standard to their local conditions. Because higher standards impose higher compliance costs if standards are set too high relative to the firm's capabilities and resources, either the firm will be forced to stop production for not complying with the standard or it will continue to produce without meeting it. In contrast, if standards are set too low, the benefits of higher standards (such as consumer safety) will be forsaken. While there are good reasons to have high safety, labor, and environmental standards, there are also good reasons why these standards are more stringent in wealthier countries. Insisting on higher standards can stymie the very development outcomes that high standards are meant to foster.

Hence, when should countries adapt the international standard to local conditions and when should they adopt it as-is? Given the inherent knowledge production, economies of scale, and network effects that standards produce, WDR 2025 will examine whether or under what conditions countries should adopt the international standard as-is, following the "one-size-fits-all" principle. *Compatibility standards* benefit from one-size-fits-all because any deviation from the standard can cause incompatibilities. For example, the TCP/IP internet protocols were developed to standardize data transmission and enable communication between different computers (Greenstein 2000; Hafner and Lyon 2006).

Problems arise when multiple standards exist, leading to "standards wars" as companies push their own standards. When network effects are strong, one standard usually wins, although it may be suboptimal (such as the QWERTY keyboard). The universal adoption of a winning standard may increase market power when one or a few unique producers provide the standard or any component needed to meet it, possibly due to the use of standard essential patents (SEPs) or other forms of intellectual property (IP) protection. Alternatively, if the costs of switching to another standard are high, multiple standards may coexist, as in the case of electrical plugs (Rogante, de Vries, and van Wessel 2022). A universal plug would reduce costs and increase trade, but replacing all outlets is too expensive compared to using adapters (David 1987; David and Greenstein 1990; Mei and Xu 2024; WorldStandards 2024). In this case, adapters and other off-ramp solutions are developed.

Measurement standards also benefit from a one-size-fits-all approach because their goal is comparability. However, some standard regimes predate globalization and have instead served as the foundation for economies and societies—making them socially, economically, and politically costly to change. As a result, these may coexist if switching to a common one is too costly. For example, despite the benefits of universally adopting the metric system, many countries still use both the metric and imperial systems, leading to significant transaction costs. In addition, there is often no agreement on the best method to measure policy outcomes like poverty, education, or health achievements (refer to box 1).

Box 1 Standards in measuring income and poverty

International standards have evolved to measure the income and consumption of households, and to determine how to identify the minimum consumption or income necessary for a household to not be classified as poor.

The first widely adopted international standard concerning poverty measurement was the dollar-a-day line introduced in the 1990 *World Development Report*. This measure has been revised several times since to account for new data, most recently to US\$2.15/day (Jolliffe and Tetteh-Baah 2024). The common usage of the international poverty line has affected performance benchmarking, setting of national and global targets, and resource allocation across countries (Crosswell 2015; Pape and Ali 2023).

Poverty measurement, however, is not one-size-fits-all. To minimize errors in poverty measurements (Grosh et al. 2022; Pu et al. 2024), governments use the best data available—which does not mean the same thing across the board (Deaton and Zaidi 2002; Giambastiani et al. 2022). As a result, poorer countries rely on consumption-based measures of poverty, while richer countries use income-based measures. The standards on income measurement are less relevant for poorer countries where a large share of the population has no formal income (Carletto, Tiberti, and Zezza 2022); conversely, the standards required to measure consumption place an enormous burden on survey respondents in richer countries, where consumption is higher and more diversified. In addition, many countries adopt direct measures of households' well-being and multidimensional poverty measurements.

Conflicting standards also exist with respect to national poverty lines, reflecting both methodological and normative decisions about what is required to escape poverty (Allen 2017; Decerf 2023; Mahrt et al. 2022). For example, national poverty line standards differ according to whether being "non-poor" requires an income sufficient to meet basic caloric needs or to afford a diversified, healthy diet. With respect to the international standard, the international poverty line is considered too low for countries not at the lowest stages of development, where a daily income level of less than US\$2.15 seems inconceivable. Though higher standards exist and are easy to apply (Jolliffe and Prydz 2021), no single standard is relevant for all countries.

Source: WDR 2025 team.

The need to adapt standards to the local context is perhaps stronger when considering *minimum standards*. Countries or regions may face very different costs of complying with the international standard, driven by differences in production capacity, citizens' preferences, and enforcement capacity (refer to box 2).

Box 2 Setting minimum standards for goods and services

How should countries set minimum standards? Figure B2.1 presents a conceptual framework. While this framework can be used for many sectors, it is tailored here for production standards. In both panels, the x-axis shows the quality in production of a product (for example, garments), while the y-axis shows how often firms can deliver a certain level of quality. The solid black curve shaped like an inverted-U represents the quality distribution in a developing country without any standards (refer to panel a) or a developed country with standards (refer to panel b). In panel a, most firms offer low-quality products or services, so the curve is centered on the low-quality (left) side of the x-axis.

Complying with voluntary standards for international markets

International standards serve two main purposes: they allow firms to demonstrate the quality of the goods and services they produce, and they help firms learn how to produce high-quality products (in this case, garments). Firms may hire consultants that teach them how to implement the standards and, if deemed relevant, also achieve certification.

Producers wishing to export need to meet the relevant standards of its potential buyer. Such standards may include voluntary standards as well as mandatory standards embedded in the technical regulations in the importing country. For instance, garment exporters may need to comply with International Social and Environmental Accreditation and Labeling Alliance (ISEAL) and International Labour Organization (ILO) standards to ensure environmental and worker safety requirements; International Organization for Standardization (ISO) management standards to ensure quality management; and ISO garment-specific standards to ensure quality in features such as color and fiber composition.

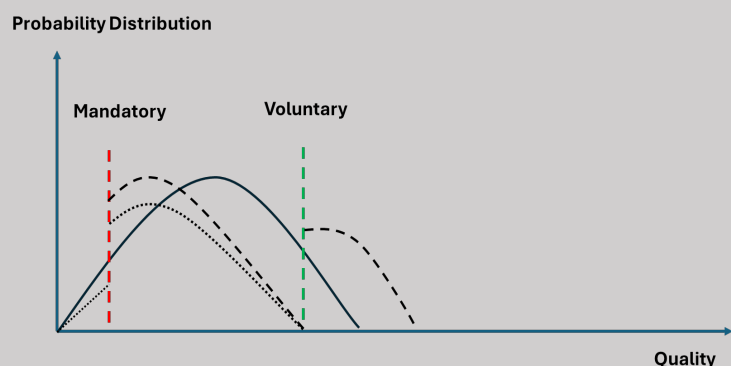
Figure B2.1 illustrates the mandatory standard as the vertical red line in panel b. From the perspective of the developing country, such a standard only applies to its firms that wish to export. As such, it can be considered voluntary and represented by the green vertical line in panel a. Note that the voluntary standard in panel a and the mandatory standard in panel b are at the same level.

Setting mandatory standards for the domestic market

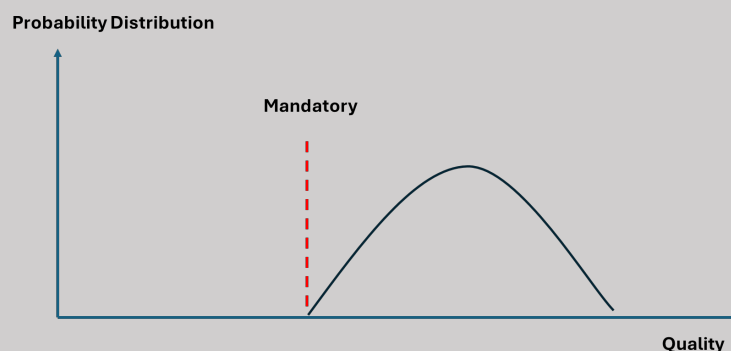
Suppose now that the government is concerned about the quality in the *domestic* garment market. It will enact regulations that “raise the floor” by imposing a minimum quality standard on all producers selling in the domestic market. Governments set its mandatory minimum quality standard by *adopting* the international standard as-is. Most garment producers would struggle to comply with this standard. Governments may instead *adapt* it and set a less stringent one as the mandatory national one (represented by the red vertical dashed line in panel a).^a

Figure B2.1 Adapting the international standard or adopting it as-is

a. Quality distribution in a developing country



b. Quality distribution in a developed country



Source: WDR 2025 team.

The dashed black curve in panel a of figure B2.1 shows how the distribution of garment quality in a developing country changes when both the voluntary international standard (the green dashed vertical line) and mandatory national standard (the red dashed vertical line) are fully enforced. After enforcing the mandatory standard, all garment producers meet or exceed this standard. However, if enforcement is weak due to lack of resources, corruption, or bribery, the quality distribution shifts to the dotted black curve, meaning that some producers still make garments below the required quality.

Setting the mandatory standard involves trade-offs. Panel b shows the quality distribution of garment producers in a developed country. Because all garments consumed in the country need to meet its mandatory standard and importers cannot be discriminated against, the government *adopts* the voluntary international standard as the mandatory national standard. This is why the red dashed line in panel b coincides with the green dashed line in panel a.

Figure B2.1 thus suggests that developing countries will *adapt* the voluntary international standard to set a less stringent mandatory national standard. In contrast, developed countries will *adopt* as-is the voluntary international standard as their mandatory one. To add empirical rigor to this conceptual finding, WDR 2025 will work with ISO, the International Electrotechnical Commission (IEC), and other international standard-setting organizations, as well as National Standards Bodies (NSBs) from several countries, to assess whether the gap between the national and international standards declines as the country's income per capita increases.

Trade leakages from one country to another when national standards differ

Figure B2.1 can also be used to explain trade *leakages* that will inevitably arise when national standards differ in different countries. Suppose that a developed country in panel b makes the mandatory emissions standards for vehicles more stringent (that is, shifts the red dashed vertical line to the right). Owners of cars that no longer meet this new national standard can either pay to retrofit their car to reduce emissions to the new standard or sell the car. While the car can no longer circulate in the developed country, it can be sold to an importer from a developing country, where it can still circulate as a second-hand car. Regulatory changes in the developed country will affect the goods traded in developing countries.

Source: WDR 2025 team.

a. To see why governments should adopt the international standard whenever possible, consider a developing country that wants to import solar lamps for rural electrification. Lamp importers may lack the technical expertise to draft the technical requirements in the procurement documents. If the international standard were enacted into regulation, importers could simply ask that imported lamps comply with the international standard without requiring detailed knowledge of the technical specifications. In addition, the higher demand for lamps complying with the international standard will push down the price because those lamps can be mass-produced. In contrast, if the country sets a national standard that differs from the international one, then lamps complying with the country's regulations could be more expensive to produce.

Authoring international standards

Many international standards are set by international standard-setting bodies such as the International Electrotechnical Commission (IEC), the International Organization for Standardization (ISO), and the International Labour Organization (ILO) through consensus among member countries (in the case of the ILO) or among delegates from countries that have paid the member annual dues (in the case of the IEC and ISO). Once countries become ISO members, they select which technical committees to join. However, because committee meetings can be costly in terms of time, and require technical expertise, financial resources, and political capital, developing countries often participate only in the committees that are most relevant to their interests. For example, developing countries producing garments will tend to participate in the ISO committees related to garment production to ensure that national producers can comply with the standard. As a result, developing countries often find themselves as "standard takers." Developed countries, in contrast, participate in almost all committees and do so more actively, thereby dominating the standard-setting process. Importing countries will also try to influence the standard to protect their local producers. Research shows that consensus-based decision-making works well for encouraging adoption, although it can also produce conflict and cause delays (Farrell and Simcoe 2012; Simcoe 2012).

Countries can sometimes take on a leadership role in authoring standards within specific sectors. In some instances, a country can also transition from the absence of standards for a product or a service to authoring its own, such as Ethiopia's teff grain quality standard. Globalization has led to the internationalization of standards. Only countries with technological prowess, state capacity, political will, and concentrated production capabilities in specific goods or services have the clout to author standards. However, this phase also presents risks such as imbalances in market power, where leading countries impose their own standards on others, potentially leading to fragmentation. Countries must navigate these dynamics carefully, choosing which standards to adopt while managing the implications for technology diffusion and international influence.

Moreover, setting standards can be costly. Therefore, governments generally prefer that standards be “open” and freely accessible to everyone. Some firms will prefer “closed” standards that require users to pay to access them. Standards also generate positive externalities. When a country adopts international accounting standards, for example, it not only improves its own financial reporting but also enhances global economic transparency and facilitates cross-border investments. Consider how ISO operates. It is not a top-down bureaucracy but a collaborative ecosystem where in principle experts from its member countries negotiate and develop standards through consensus building. In practice, of course, many of its member countries simply lack the experts who can meaningfully participate in these meetings.

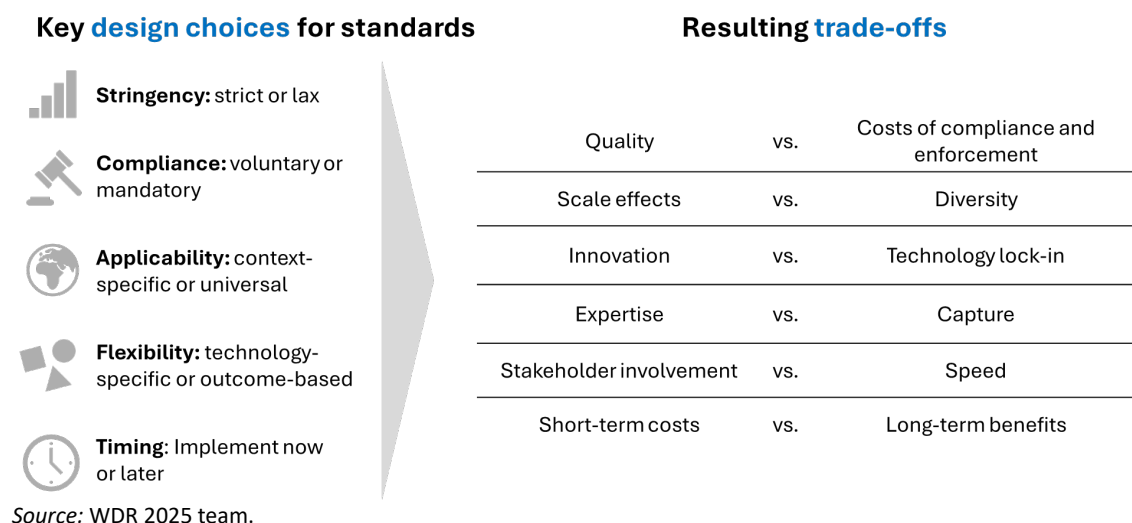
Political aspirations sometimes dominate discussions on the setting of standards, reflecting a nation's ambition to shape the global rules of the game, such as the United States' push for 5G standards formulated by its firms, or the European Union's General Data Protection Regulation (GDPR). These initiatives were not only about creating technical specifications; they were assertions of geopolitical influence and attempts to set the terms of engagement for the digital age. But authorship is not limited to global superpowers. Smaller nations and even nonstate actors can play crucial roles in standard setting, particularly in emerging fields where the rules are still being written. Estonia's digital governance standards and the Linux Foundation's open-source software protocols demonstrate how focused efforts can have outsized impacts on global norms (World Bank 2021). Understanding this journey from adoption to participating in standard setting becomes crucial as countries navigate an increasingly complex world. It is not just about technical specifications or economic efficiencies. It is about how nations assert their identity, pursue their economic interests, and project their vision for the future onto the global stage.

WDR 2025 will provide an overview of national, regional, and international standard-setting organizations, detailing their roles and structures. It will also work with the international standard-setting organizations like ISO, the World Trade Organization (WTO), and the Sanitary and Phytosanitary (SPS) Committee and Technical Barriers to Trade (TBT) Committee of the WTO to understand the capacity building that developing countries require to support their participation in these committees, building on the experiences of successful developed and developing countries.

Trade-offs in setting standards

The design choices for setting a standard entail a range of trade-offs (shown in figure 3). Key design choices include how stringent a standard should be, whether compliance is voluntary or mandatory, whether it is universally applicable or specific to a context, and whether it is outcome-based (specifying, for example, a level of energy efficiency) or technology-specific (stating, for instance, that buildings must have double-glass windows). Another key design choice concerns the timing of the standard—whether to introduce it early, while technologies are still evolving, or whether to wait until the market has converged on a dominant or mature solution. The design choices are linked to a set of trade-offs. Achieving scale may reduce diversity. Having subject matter experts lead the development of standards may bring in expertise at a greater risk of capture. Wide stakeholder involvement may improve the quality of the standard and uptake but may also delay the standards development process. Trade-offs differ by the areas of application of standards, a country's level of development, and societal preferences.

Figure 3 Design choices on standards and trade-offs



A key trade-off lies between quality and the cost of compliance or enforcement. Higher quality standards impose higher costs. If the government sets the standard too low, the quality will be too low; if it sets the standard too high, products will be too expensive. The higher the standard, therefore, the greater the exclusion of consumers (due to high prices) and producers (due to high compliance costs). Governments should, therefore, assess the competing demands of consumers and producers. Consumers may have quality concerns and be willing to pay for higher quality, thus demanding a more stringent standard. In contrast, producers may want to use cheaper but lower-quality inputs. Lacking expertise, governments, through their regulatory agency, may defer to the industry, potentially resulting in capture and a less stringent standard.

The costs and benefits of setting standards may occur over different time horizons and possibly generations, with compliance costs incurred immediately and benefits realized in the longer term, such as in the case of greenhouse gas emission reductions and the impacts of climate change. In addition, standards will modify the incentives faced by consumers and providers. Demanding higher quality will provide incentives for producers to invest in better inputs and processes that allow for higher-quality products at lower prices and induce consumers to demand higher quality; stricter standards will require developing better monitoring and enforcing technology, implying possibly significant investment costs but also making it easier to enforce standards in the future.

As a result, the government may set standards that exceed the currently optimal level, laying the basis for higher quality and aggregate welfare. These dynamics highlight the fundamental way in which standards are at the heart of the development process (de Vries and Verhagen 2016).

The design choices on standards can, however, create risks. Standards, for example, can be used as barriers to trade. In 2000, Zambia mandated that sugar sold within its borders be fortified with Vitamin A. Zambia Sugar, the dominant domestic producer, supported this move. Cheaper producers from neighboring countries and small producers in the country lacked the means to fortify sugar with Vitamin A and, as a result, were adversely affected because they could no longer sell sugar in Zambia. Consumer prices for domestic sugar rose in Zambia, with Zambia Sugar raising its prices by 14 percent in 2012 (World Bank 2017).

Similarly, the increasing use of geolocation devices to recover stolen vehicles in South Africa in the 1990s led insurance companies to demand industry standards for stolen vehicle recovery (SVR) services. The trade association VESA established performance-based criteria that required SVR service companies to have been in operation for at least one year, completed 3,000 unit installations, and achieved 100 successful recoveries (World Bank 2017). These standards and the fact that insurance companies offered premium discounts exclusively to car owners using VESA-approved systems created substantial barriers to entry for new entrants.

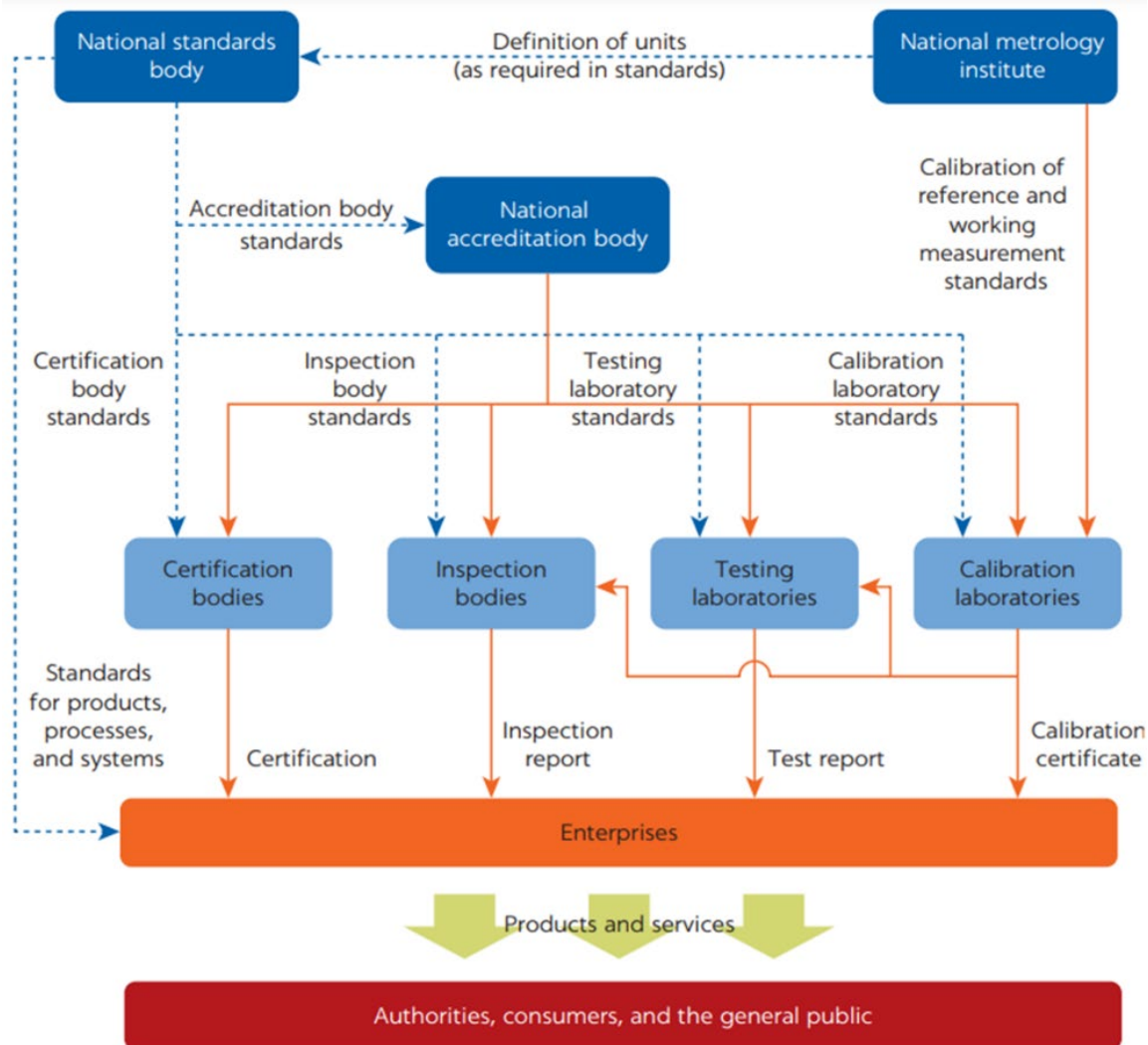
Standards may also have unintended consequences. Consider the case of how building codes affect equality of access to restrooms for men and women in business facilities. Huh et al. (2019) found that building codes such as the Uniform Plumbing Code (UPC) implemented gender equality based on the number of toilets (fixtures) rather than waiting times influenced by gender-specific needs, which led to longer wait times for women.

In some areas, such as trade, the proliferation of standards can create a “spaghetti bowl effect,” with negative effects on trade relationships (Bhagwati and Krueger 1995). For instance, in 1998, the European Commission (EC) announced a proposal to harmonize standards to regulate aflatoxin, a family of poisonous molds that have been found to cause cancer and other mutations. The standard was implemented in 2002. Compared to the levels before the EU harmonization, the new EU standard decreased African exports of cereals, dried fruits, and nuts by 64 percent, equivalent to a loss of US\$670 million, Otsuki, Wilson, and Sewadeh (2001) found. This loss was particularly large when compared to a scenario where standards were set according to the guidelines of the Codex Alimentarius, the preeminent international food standards-setting body. The study also showed that the new EU standard would reduce cancer deaths per year by only two per billion people compared to a standard set by Codex.

How to implement and enforce standards?

In practice, the implementation of standards and the enforcement of technical regulations are supported by a quality infrastructure (QI) system (refer to figure 4). This system comprises public and private organizations together with the relevant legal and regulatory framework and practices needed to support and enhance the quality, safety, and environmental soundness of goods, services, and processes (INetQI 2021). It relies on five elements: *metrology*; *standardization*; *conformity assessment* (the process of evaluating whether a product, service, or system meets the requirements of a standard via testing, inspection, and certification); *accreditation* (the formal recognition that an organization can perform conformity assessment); and *market surveillance* (the process of monitoring products and services in the market to ensure they comply with regulations and standards).

Figure 4 The national quality infrastructure (QI) system



Source: Kellermann 2019.

QI systems are typically designed to ensure product quality, and are thus less relevant for compatibility standards because producers self-enforce them to capitalize on the benefits of network effects. Nonetheless, the WDR 2025 argues that many of the elements and principles of a well-functioning QI system also apply to basic services such as education and health, finance, and public digital infrastructure. In education, for example, universities need to be accredited before they are allowed to issue degree certificates that confirm students' achievement of minimum education standards. For production, market surveillance ensures that products in the market carry legitimate safety labels resulting from proper testing by competent bodies in line with applicable standards.

The five elements of a QI system are essential to ensure quality and prevent conflicts of interest. For instance, if the same body sets standards and handles certification, it might create standards with the sole intention of generating revenue from certification, especially for mandatory standards. Similarly, accreditation should be separate from conformity assessment, and the party/parties assessing conformity must be impartial. In developing countries, separating these roles can be difficult because different QI services generate varying amounts of revenue, often requiring cross-subsidization.

While voluntary and mandatory standards depend on the same elements of a QI system, the body authorized to perform each function may change. For example, in many countries, ISO 9001 is adopted by organizations to improve their internal systems and signal quality to other market participants. While adoption of the standard is voluntary, firms may choose to use third-party certification to demonstrate their conformity with the standard. For certain mandatory standards, governments typically specify which recognized conformity assessment bodies need to be involved to demonstrate conformity.

While the outlined QI elements are typically present, national QI systems have evolved in different ways, reflecting a country's history, economic structure, state capacity, and political-administrative traditions. The Report will shed light on three interdependent trade-offs along which QI systems differ:

1. *Coordination versus competition.* Should one or many National Standards Bodies (NSBs) manage the standards development process? An NSB may facilitate coordination and ensure coherence, but it may be slower to respond to innovation and market demand, especially if specialized knowledge is required. In contrast, the QI system in the United States relies on more than 600 standards development organizations (SDOs) with direct links to the industry. The American National Standards Institute (ANSI) operates a recognition scheme for SDOs to ensure that good practices are followed as standards are developed.
2. *Private versus public bodies.* Should the functions in a QI system be carried out by the government, the private sector, or a public-private partnership (PPP)? The answer largely depends on the QI service being delivered and the development level of the country. For example, product certification is typically done by private bodies in high-income countries but may initially require more government involvement in developing countries, from helping finance the capital equipment for testing labs to creating the public infrastructure for online portals integrating traceability modules to Laboratory Information Management Systems (LIMS), and so on.
3. *Precaution versus correction.* Should regulated products and services be assessed and certified *before* they are placed on the market or *after*? The answer depends on the risk involved with non-compliance. Many high-income countries have moved to a laissez-faire approach that puts less emphasis on pre-market assessments and relies more on post-market surveillance and legal action. This more business-friendly environment requires well-functioning surveillance capacities and an effective legal framework—both of which may be deficient in developing countries.

A well-functioning QI system relies on an effective interplay of private and public sector institutions. In many QI areas like metrology, a QI system depends on the capacity of governments and their public institutions (Kellerman 2019; World Bank 1997). Capable institutions set standards informed by accurate assessments of their cost and benefits to sectors, reducing regulatory burden (Kirkpatrick and Parker 2024; Ladegaard 2005), and improving compliance by regulated firms (OECD 2014; Malesky and Taussig 2019). WDR 2025 will document how state capacity can be improved by drawing on recent literature from procurement agencies, tax offices, and more (Best, Hjort, and Szakonyi 2023; Khan, Khwaja, and Olken 2016). For example, regulators may be required to have sector-specific expertise to ensure that regulated entities, such as banks, comply with the existing regulatory framework (Anginer et al. 2019); or cool-off periods could be mandated to slow down revolving doors and make regulations less likely to favor specific firms (Dal Bó 2006).

Part III: Standards *for* Development

Successful countries have used standards to propel development outcomes, building capacities to comply with international standards, and ultimately to develop new standards. Countries such as the Republic of Korea and, more recently, China have shifted from marginally participating in international standards-setting organizations such as ISO or the IEC to chairing some of their working groups and developing an increasing number of standards. This process occurred not only in the areas of production and technology, but also in education (Korea) and the environment (China). WDR 2025 will examine how developing countries can replicate these experiences of successful countries in different contexts, despite often having limited financial and human resources to adopt and enforce international standards and build the necessary institutions and QI systems. It will also suggest what external technical and financial resources governments may need to meet these challenges.

Speeding up these transitions to boost development comes with risks due to the changing environment for adopting international standards. Geopolitical tensions and political fragmentation are rising, along with demands for more environmentally sustainable production in developing countries. New technologies are reshaping production and lowering information costs, but also increasing competition to set the dominant standards. The return to industrial policies is leading to more measures supporting national industries and expanding areas that are considered sensitive or critical. In this context, the multilateral system in 2025 is very different from that of 2015 or 2005, creating significant risks for development.

Part III will provide insights and lessons on how transitioning across standards can advance development through four key channels: a better society (improving education and health care); a better economy (improving production, trade and investment, technology, and finance); a better environment (local and global); and better governance (state capacity and policy).

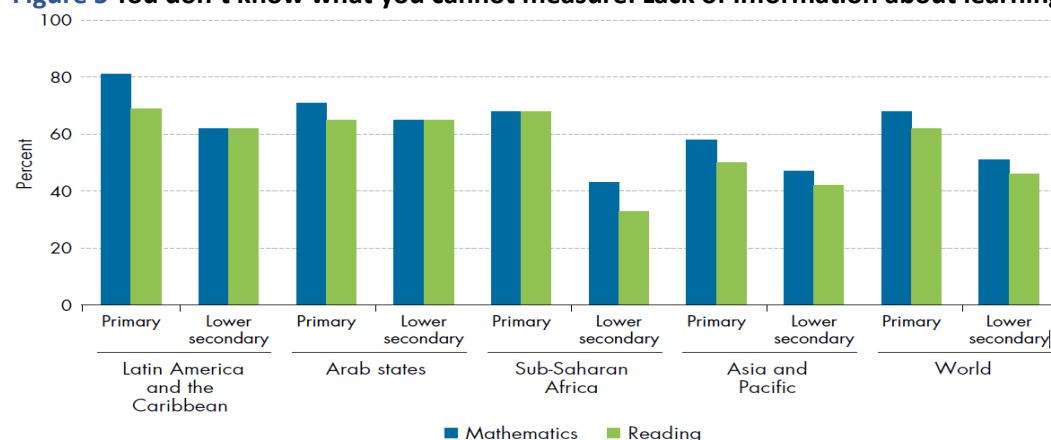
Standards for a Better Society

The chapter on Standards for a Better Society will explain how standards can enhance basic services such as education and health care. It also will cover standards for skills certification at both national and international levels. Developing countries often struggle with limited access to these essential services, which also vary in quality and consistency. Standards can help improve the quality of these services. Moving from a lack of standards to adopting them, and then setting national standards is crucial to identifying gaps and guiding policies. Basic services are non-tradable and may require careful consideration of the specific context in which they operate; hence, adapting international standards can further enhance service quality.

Education

Education standards track such aspects as infrastructure, enrollment rates, and pupil-teacher ratios, and have helped significantly improve school enrollment rates around the world (UNESCO 2023). Despite nearly universal primary enrollment in low-income countries and lower-middle-income countries, learning outcomes remain a challenge, with many children failing to meet minimum reading and math proficiency standards. Moreover, many countries lack standardized systems for monitoring progress related to learning outcomes. Significant gaps exist in measuring progress toward the Sustainable Development Goals (SDGs) related to learning outcomes at the end of primary and lower secondary school, particularly in Sub-Saharan Africa and the Asia and Pacific region (refer to figure 5). Measuring standards is crucial because education cannot be improved if learning outcomes are not tracked.

Figure 5 You don't know what you cannot measure: Lack of information about learning outcomes



Source: World Bank 2018.

Note: The figure shows the percentage of countries with data to monitor progress in reading and mathematics proficiency levels at the end of primary and lower secondary school.

International measurement standards like the Programme for International Student Assessment (PISA), the Progress in International Reading Literacy Study (PIRLS), and the Trends in International Mathematics and Science Study (TIMSS) assess student learning. They provide data that can guide educational reforms and help governments decide where to invest in education. However, participating in these assessments can be expensive due to the high costs of sampling, translating, and administering the test. The test may not always address the specific learning challenges of developing countries because the assessment framework lacks cultural relevance and does not align with the student's specific learning challenges and needs.

The chapter will examine measurement standards for learning outcomes used nationally and globally, highlighting the challenges faced by developing countries in meeting their own national standards and offer policy recommendations for implementing these standards in low- and middle-income countries.

Health care

The global demand for health care workers is projected to rise significantly by 2030, potentially leading to a shortage of millions of professionals (Liu et al. 2017). However, many countries underutilize their existing health care providers. For instance, approximately 75 percent of health providers in Nigeria see fewer than two patients per day (Das and Do 2023). While 91 percent of the global population lives within an hour of a health care facility (Weiss et al. 2020), many patients are still misdiagnosed, especially in developing countries. Health care workers in seven African lower-middle-income countries had an accuracy rate of only one-third to three-quarters in diagnosing conditions (Kieny et al. 2018).

Standardized guidelines can enhance health care quality by setting minimum requirements and ensuring workforce competency (Sinsky et al. 2021). A good example highlighting the benefits of standardization is a study in three Kenyan counties to test the impact of a minimum quality standard (MQS) in health care facilities (Bedoya et al. 2019). Government inspectors visited these facilities to check their conditions and either recommended that they be closed or set a timeline for improvements. The enforcement of standards resulted in better quality in public and private health care facilities, increased trust in public health care, and improved patient safety and health care delivery while maintaining access to care.

The chapter will outline the various standards in health care inputs, processes, and outcomes and discuss the benefits and potential drawbacks of a one-size-fits-all approach to the quality of care. It will present global data on density norms, medical qualifications, and health care infrastructure, and analyze lessons learned from applying MQS in developing countries, ultimately offering policy recommendations to address the challenges faced in low-income and middle-income contexts.

Standards for a Better Economy

Standards are essential for improving production quality, fostering innovation, ensuring access to finance, and promoting economic growth. The chapter on Standards for a Better Economy will examine how standards can help upgrade production in developing countries and will investigate the impact of stricter international standards on exports. Many low-income countries struggle to transition from an absence of standards to adopting standards, as compliance with international standards is becoming increasingly difficult with the proliferation of new and stricter export-related standards. The chapter will also examine the role of standards in improving domestic quality and labor relations; how to build a well-functioning QI system; how to boost exports in a world with increasingly complex trade-related standards; the use of foreign direct investment (FDI) to spread knowledge; the importance of technology standards; and finally the role of standards in ensuring financial stability and access to finance.

Production and quality

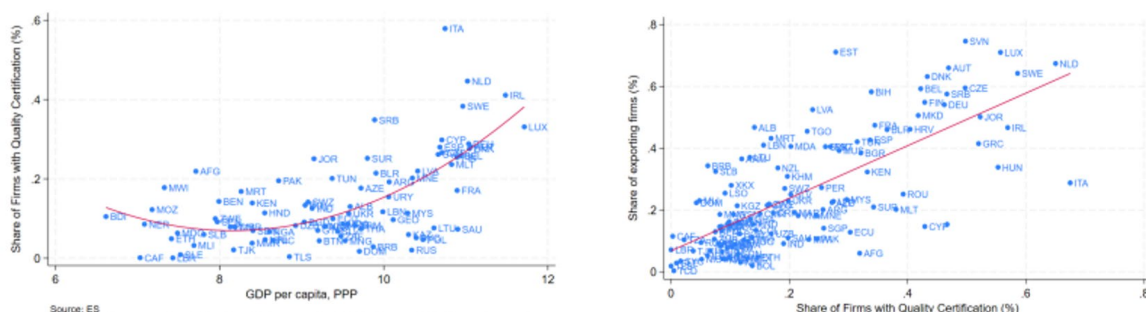
Lack of standardization can result in excessive product variety and inconsistent quality, preventing economies of scale (Dunlavy 2024). Product standardization can change production so that workers specialize in pursuing a few tasks within the firm, resulting in higher productivity and firm growth (Holmes and Stevens 2014). For example, analysis of a sample of carpentry, grain-milling, and welding microenterprises in Uganda showed that firms that produce standardized products exhibited greater labor specialization and tended to be larger and more modern (Bassi et al. 2023). Adopting international standards provides access to codified knowledge, as evidenced by THEDAY, a woman-led jam manufacturer in Ethiopia that enhanced productivity by adopting the ISO 22000 food safety standard (World Bank 2024).

The challenge for developing countries

Firms in developing countries must meet increasing environmental sustainability and social standards to access export markets. The importing country regulators set some of these international standards, while private buyers set others to meet their consumer and social responsibility demands. Developing countries have limited ability to influence these international and private standards. This poses a risk if firms cannot quickly adapt their policies and institutions to adopt these standards, which can result in lost export opportunities. As a result, firms in developing countries are less likely to adopt voluntary international standards (refer to figure 6). Furthermore, firms are more likely to implement technical regulations and mandatory standards that are less stringent (refer to figure 7).

Figure 6 Firms in developing countries are less likely to adopt voluntary international standards

- a. Share of firms certified as per ISO 9001 and GDP per capita b. Share of firms certified as per ISO 9001 by exporting status

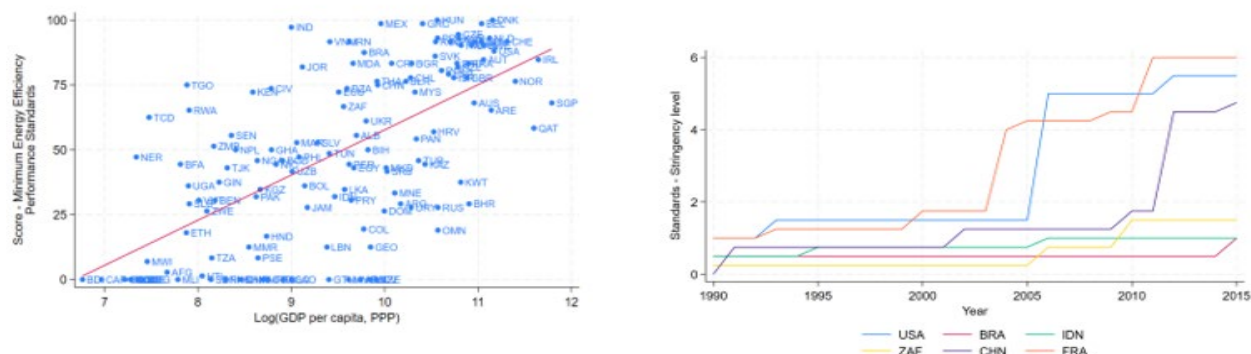


Source: WDR 2025 team using the 2024 World Bank Enterprise Survey (World Bank Group 2024).

Note: Data labels use ISO country codes.

Figure 7 Firms are more likely to implement technical regulations and mandatory standards that are less stringent

- a. Minimum energy-efficiency performance standards b. Stringency



Source: OECD 2024.

Note: In panel a, scores refer to the average score for the country. In panel b, stringency levels range from 0 to 6, with 0 being least stringent and 6 being most stringent. Data labels use ISO country codes.

Labor relations

Member states of the ILO are encouraged to ratify ILO conventions, but after ratification, many states—especially developing countries—struggle to enforce the standards set in the conventions. The chapter will consider describing the setting and enforcement of labor standards around the world using the Report's conceptual framework.

Building a well-functioning QI system

Developing countries set up QI systems to address both domestic needs and external demands. Technological progress requires accurate and reliable assessments, raising questions about public versus private investment in quality infrastructure and what areas to prioritize. Third-party certification systems (TPCS) can strengthen weak state capacity. Setting up a solid business model, and properly sequencing reforms and infrastructure investments, are essential to developing effective QI institutions. This is critical given that emerging standards may challenge QI institutions in their current set-up as they impose more stringent traceability requirements, such as recording the amount of greenhouse gas (GHG) emissions in the production, providing proof that production did not contribute to deforestation (the European Union Deforestation Regulation, EUDR), or demonstrating that production is not using child or forced labor. The chapter will highlight successful cases in developing the QI system to accelerate the standards transition.

Trade

A number of non-tariff measures (NTMs), including sanitary and phytosanitary (SPS) measures and technical barriers to trade (TBT), have gradually replaced tariffs over the past 30 years. These measures govern product characteristics and production processes, creating a complex "spaghetti bowl" of requirements for developing countries. While the impact of SPS and TBT measures on trade is mixed, new standards increase compliance costs, although they can also improve product quality and consumer information.

The chapter will assess whether voluntary and mandatory standards related to trade are becoming more stringent and their impact on developing country exports. The chapter will map the evolution of voluntary and mandatory standards related to trade over time to quantify the nature of this "spaghetti bowl." It will also review the impact of these standards on exports from developing countries and will explore whether there is evidence that exports are being diverted to less stringent destinations. This stringency creates a trade-off between compliance costs and export diversification. Firms may need to focus their efforts on a single market, leaving them vulnerable to demand shocks, regulatory changes, or policy shifts in that market. Furthermore, importing countries are increasing stringency through the growing use of due diligence standards. At the same time, conformity costs are shifting from importing to exporting countries, increasing compliance costs borne by exporting countries and adding additional pressure on their QI systems. Finally, the chapter will examine the international community's efforts to harmonize standards aimed at balancing quality and environmental goals with market access and equity. The Report will explore how these efforts affect developing countries.

Investment

FDI is key to diffusing technologies and international standards. Adopting international standards can help in attracting FDI and accelerating this diffusion. For example, Iacovone et al. (2011) find that the effects of firms in Mexico selling to Walmart differ depending on firm size and level of quality of production. Zeng and Eastin (2012) find that FDI has a positive effect on the adoption of ISO 14001 sustainability standards by the least developed countries.¹ At the same time, the adoption of standards by various countries has contributed to attracting FDI in various countries. The chapter will explore policies that developing countries could implement to attract FDI and increase the transfer of technology standards from foreign to domestic firms. Specifically, the discussion will review the evidence of how standards have contributed to attracting FDI in various countries and in what ways standards have driven more sophisticated production, fostering economic upgrading. Finally, the section will look at how standards are expected to evolve for investment and the implications for the participation of developing countries in global supply chains.

Technology

Technology standards ensure interoperability, drive innovation, and reduce inefficiencies across systems and markets. Recognizing this, industries establish thousands of interoperability standards without any government involvement. As part of the innovation process, firms may simultaneously develop competing technologies; a "standards war" can ensue as each tries to establish its proprietary technology as the dominant one. Notable examples include VHS versus Betamax, Blu-ray versus HD DVD, and, more recently, rivalries in 5G infrastructure. By being standard-takers more than standard-setters, developing countries can be especially vulnerable to the outcome of these wars. Adopting a standard that will no longer be used or one that is incompatible can lock these countries into outdated technologies, stifling productivity and competitiveness in global markets. Conversely, adopting widely accepted superior technologies can serve as a technological leapfrog, for example, with the widespread adoption of mobile phones.

Interoperability standards help technologies work together across different systems and devices, reducing costs and enabling wider adoption. However, these standards often rely on Standards-Essential Patents (SEPs), necessary for implementing technical standards. Licensing these patents under fair, reasonable, and non-discriminatory (FRAND) terms is crucial to lowering barriers to adoption (Bekkers 2021). Access to affordable SEP licenses for developing countries can encourage adoption and promote innovation. However, SEPs can also increase the market power of large tech firms. Rising geopolitical tensions may fragment the global standards system, forcing developing countries to choose between different standards.

The chapter will evaluate the impact and responses to these risks. It will examine how different models of interoperability and technology standards influence the spread of technologies in developing countries. It will also discuss the best timing to set standards, the balance between supporting innovation and protecting intellectual property rights (IPRs), potential competition risks with SEPs, the speed of technology adoption, and the importance of open versus proprietary systems in developing countries, especially in digital technologies.

Finance

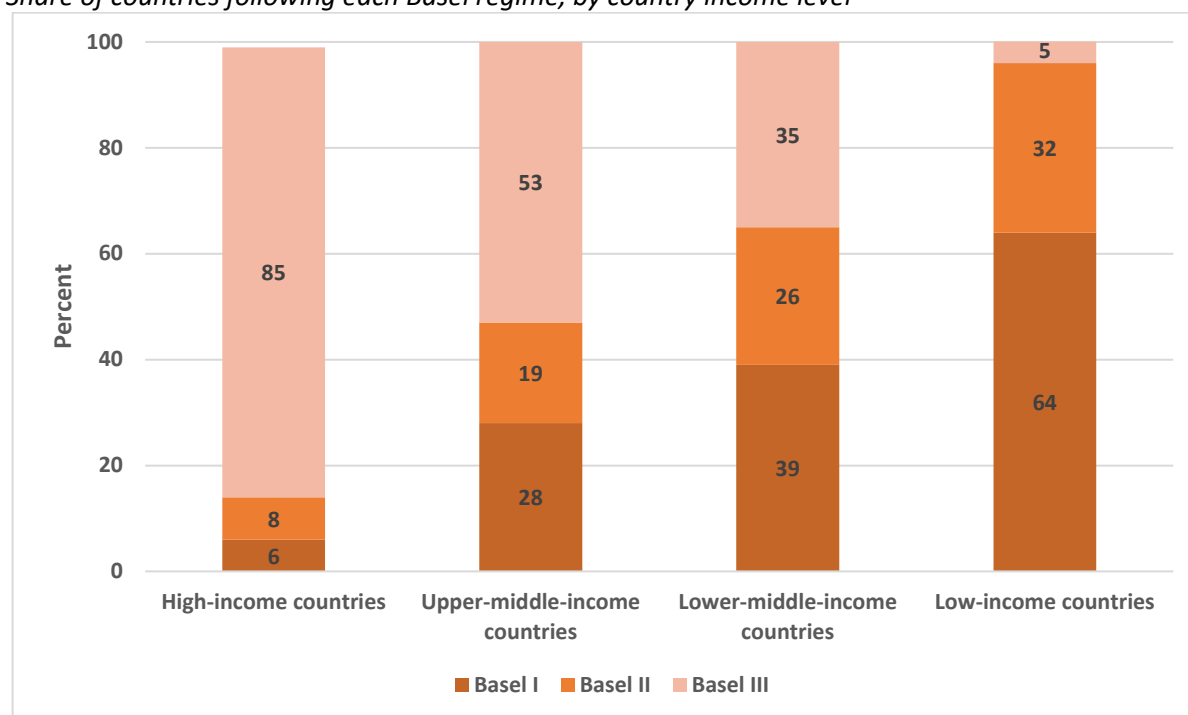
Bank capital regulation

Bank capital, including shareholders' equity and retained earnings, provides a cushion during distress and curbs risk-taking. Because it is costly for banks to hold capital, they may hold insufficient amounts, relying instead on safety nets like deposit insurance or bailouts during crises (Calomiris 2012). From Basel I to Basel III, the Basel Committee on Banking Supervision (BCBS) has used a risk-based tiered approach to set international banking standards. The goal is to prevent banks from holding insufficient capital and avoid systemic risks.

The adoption of Basel standards varies according to the country's income level (refer to figure 8). This lack of uniformity can influence how international banks allocate their lending and risk-taking across jurisdictions. For example, stricter US capital regulation reduced lending by large US global banks in foreign markets, Berrospide et al. (2016) demonstrate. A study examining business lending by 155 banks to firms across 16 countries in Europe and Central Asia found that stricter domestic capital requirements are associated with reduced lending standards abroad (Ongena, Popov, and Udell 2013). This finding suggests that foreign banks exploit more lenient host-country regulations to offset constraints on high-risk, high-return lending at home. Differences in capital regulation standards influence how banks allocate lending and risk-taking across jurisdictions. Stricter regulations can reduce lending abroad and affect borrower targeting.

The incentive to participate in Basel standards has served as a powerful motivation for countries to transition to standard adoption. Ghana, Rwanda, and Kenya, for example, have advocated for and implemented Basel II and III international banking standards, despite not being members of the Basel Committee. Pakistan has also achieved a high level of adoption of Basel II and III standards. This process was driven by a combination of factors, including a policy to promote financial services, regulators seeking to implement best practices, and banks championing implementation as they internationalized.

Figure 8 The adoption of Basel standards varies according to the country's income level
Share of countries following each Basel regime, by country income level



Source: World Bank 2016 Bank Regulation and Supervision Survey (BRSS).

Note: Based on data from 156 countries.

Sustainable finance

Financial sector regulators have developed sustainable lending guidelines (taxonomies) to shift capital away from carbon-intense activities into sustainable projects. These policies include mandatory guidelines for underwriting procedures to channel financing to green activities (such as more burdensome procedures for brown projects and higher capital requirements for loans with greater exposure to climate risks). Evidence suggests that green lending guidelines have successfully promoted sustainable activities in some developing countries. Mandatory guidelines in China, which required banks to use stricter underwriting procedures for firms with major environmental and social risks, have led to less lending to such firms, reducing their emissions (Fan et al. 2021). In Brazil, making rural credit conditional on proof of compliance with environmental, social, and governance (ESG) regulations has led to less deforestation in the Amazon (Assunção et al. 2020).

Green lending guidelines, however, may not work if they leave room for leakages: for example, if guidelines are applied only to large banks, as in the case of Brazil. While climate-related capital requirements for large banks forced them to reallocate their lending away from brown sectors, smaller banks, which were not subject to the requirements, expanded their lending to brown sectors. As a result, the policy had no substantial impact on greenhouse gas emissions (Miguel, Pedraza, and Ruiz-Ortega 2024). The chapter will describe these efforts to enable green finance via standards such as taxonomies and provide recommendations based on successful case studies.

Financial inclusion

The chapter will describe changes in regulations, including Know Your Customer (KYC) requirements, that have enabled financial inclusion. It will discuss the technology standards that have facilitated digital payments as part of the digital public infrastructure. The discussion will also explore data compatibility and privacy standards that support open finance. Furthermore, the discussion will provide a detailed

examination of bank capital standards, analyzing their overall impact on financial stability and access to finance and their timing in relation to financial crises. It will investigate how different countries have successfully implemented more stringent standards over time. The section will also address standards for sustainable finance and their role in shaping flows toward specific countries and projects aimed at mitigation or adaptation. It will document regulatory actions related to sustainable finance and offer lessons for developing countries as they work to implement these measures. Finally, the discussion will examine the role of standards in carbon offset markets, using Colombia as a case study.

Standards for a Better Environment

The chapter on Standards for a Better Environment will begin with an overview of environmental standards in practice, covering different types across countries, changes in their stringency over time, and the processes by which these standards are set and enforced. This discussion will include stylized facts generated from the global database on environmental standards and economic development, such as the relationship between GDP and standard setting across countries, as well as real-world examples. The chapter will then delve into select representative standards to evaluate their impacts on the environment and economic development. These will include standards related to local environmental issues, such as water and air quality, and standards related to the global environmental challenge, in terms of both mitigation and adaptation (such as emission standards, waste management standards, and passive and active cooling standards). Finally, the chapter will draw lessons from applying environmental standards in various contexts and offer policy implications to address the critical challenges associated with environmental standards.

Water quality, fuel economy, waste management, and cooling technology standards are briefly described next. The chapter will present a more systematic discussion of the effective setting and impact of environmental standards for clean air and energy and to encourage adaptation to climate change, focusing on improvements in cooling technologies (both passive and active).

Water quality standards

Water quality standards aim to protect public health and ecosystems by managing wastewater. The World Health Organization (WHO) sets global guidelines that many countries adapt to their specific needs. These standards are typically enforced through discharge permits and subsidies for wastewater treatment plants. Effective enforcement and complementary policies, like subsidies and pollution monitoring, are essential for achieving water quality.

Fuel economy standards

Fuel economy standards reduce greenhouse gas emissions by improving vehicle efficiency. These standards vary by country and aim to balance economic costs with environmental benefits. Effective enforcement can lead to significant emission reductions, though unintended consequences, such as pollution leakage, may occur. Fuel economy standards must be well-designed to avoid incentivizing the production of larger, less efficient vehicles. How should developing countries consider fuel economy standards amidst the shift from internal combustion engines to electric vehicles?

Waste management standards

With municipal solid waste projected to increase significantly, waste management standards are essential. These standards, like the Basel Convention and EU Waste Framework Directive, regulate hazardous waste movement and promote waste hierarchy: prevention, reuse, recycling, recovery, and disposal. Efficient systems for waste design, collection, recycling, and disposal are crucial for managing the rising waste generation.

Adaptation to climate change: Cooling technology

As global temperatures rise, the demand for cooling technology will increase as a key climate adaptation strategy. Minimum energy performance standards (MEPS) for cooling products can improve efficiency and reduce emissions. In the United States, a series of MEPS applied to domestic refrigerators over a forty-year period led to a 75 percent reduction in electricity consumption and a 50 percent reduction in capital cost (UNEP 2023). At present, however, while 115 countries have established MEPS for cooling and refrigeration technologies, in Asia, Oceania, and Africa, 40 percent to 60 percent of countries do not have MEPS that cover cooling or refrigeration appliances (UNEP 2023). Implementing and updating MEPS can phase out outdated cooling technologies and promote more efficient models. Developing countries must also prioritize passive cooling standards (embedded in building materials and codes), as well as urban planning standards that integrate nature-based options to adapt to increasing heat stress.

Standards for Better Governance

The capacity of governments to implement policies depends on the quality of their public institutions, which in turn depend on organizational assets such as personnel, financial resources, and information systems and how these are managed. There is growing empirical evidence that improvements in management practices associated with these organizational dimensions are linked to better performance in private and public sector organizations (Bloom and Reenen 2006; Scur et al. 2024; Sousa and Voss 2002).

The chapter on Standards for Better Governance will focus on three core management systems common across all governments and their institutions: (1) human resources management (HRM); (2) public financial management (PFM); and (3) public procurement systems. There is substantial evidence that management practices associated with each one of these dimensions can improve capacity for policy implementation.² There is also widespread acknowledgement of the damaging consequences of weak public procurement systems on cost, quality, and corruption (Abdou et. al. 2022; OECD 2016). Therefore, improvements in the quality of these three core management systems can increase public institutions' overall capacity to implement policies.

Management systems involve management principles, practices, and standards. *Management principles* are the set of broad objectives that guide the specific design of management standards (such as integrity, transparency, accountability). *Management standards* are the rules and laws that support principles and specify what practices should or should not occur. Finally, *management practices* are the de facto implementation of management standards and thus relate to whether they are enforced (Grindle 2012).

Human Resources Management (HRM)

In general, management principles such as meritocracy, independence, and accountability have been associated with improvements in institutional capacity. The chapter will provide a data-driven assessment of management practices worldwide, building on the work produced by the Bureaucracy Lab, and a specific case study using administrative data on personnel.

Public Financial Management (PFM)

In budgeting, evidence suggests that the introduction of standards that promote budgetary transparency can improve the efficiency of expenditure and reduce corruption (Chen and Neshkova 2020). The chapter will provide a case study of how standards for green procurement and budgeting can be set, alongside implementation challenges that should be considered.

Public procurement systems

Public procurement faces multiple integrity risks across all stages of the procurement process, from the needs assessment over the bidding phase to the contract execution and payment. The chapter will examine evidence from the World Bank's procurement standards in its projects in developing countries as well as how standards embedded in new technologies (from block chain to e-procurement) can enhance the integrity of public procurement.

Setting and enforcing standards

Whether it is a national accreditation body, inspection agency, or regulatory institution, public institutions enforcing standards need the personnel and financial resources to be effective. The chapter will explore which management standards best equip institutions to hire regulators with relevant expertise, increasing their ability to ensure compliance (Anginer et al. 2019). In addition, the chapter will assess the process for setting public standards. There is growing evidence that participatory approaches to the setting of standards, through consultation with stakeholders and invitation for comments, can not only enhance the design of regulations but also ensure compliance (Ladegaard and Jacobs 2010; OECD 2014). The chapter will build on available data such as Indicators of Regulatory Policy and Governance (OECD 2022) to explore how improving the accountability and transparency of public institutions that set standards can ensure that they contribute to development. Finally, the chapter will describe standards for public data transparency and compatibility.

PART IV: Policy Implications

Part II of WDR 2025 emphasizes that the process of development involves transitioning from the *absence* of standards to the *adoption* of an international standard; for some countries, the transition is from *adoption* to participation in the development of the international standard (*authoring*) or from the *absence* of standards to directly *authoring* standards. Part III provides evidence from different countries and sectors for these transitions.

Part IV will provide policy implications to assist developing countries in navigating the design choices for setting standards and managing a variety of trade-offs they encounter. It will explore the factors that shape these decisions: namely, the country's level of economic development, its technical expertise, financial resources, and state capacity. In doing so, Part IV will address three key questions: (1) what is the political economy of international standard setting (focusing on ISO and the IEC) and domestic enforcement of standards; (2) what policies and institutions can speed up these transitions; and (3) how can countries manage the risks associated with these transitions, especially the risk of “decoupling” as developed countries try to impose their standards on others.

The world is experiencing significant transformations and challenges, including increasing climate impacts, rapid technological advancements, growing inequality, complexities of geopolitics, and rising debt levels. Standardization can help address these challenges and meet future needs.

Appendix A: Consultations, Timeline, and WDR 2025 Team

Consultations and timeline

Since the topic for the *World Development Report 2025* was announced, the team has held internal consultations within the World Bank Group with Global Practices, corporate units, and technical colleagues and preliminary engagements with a range of external stakeholders (refer to appendix D).

The WDR will also benefit from the guidance and advice of an Academic Advisory Committee and consultations with other academics (refer to appendix E), as well as a High-Level Advisory Panel (refer to appendix F).

To delve further into topics covered by the WDR, the team has commissioned a series of background papers and country case studies (refer to appendix G).

WDR 2025 will be published in late 2025.

The WDR 2025 team

Xavier Giné is the Director for the Report. Professor Devesh Kapur from the Johns Hopkins University is the Lead Academic. Laurent Porte served as Task Team Leader from July 2024 to January 2025. Joyce Ibrahim has served as Task Team Leader since January 2025. The core team comprises Claudia Berg, Milo Bianchi, Miriam Bruhn, Xavier Cirera, Siddharth Dixit, Philip Grinstead, Galileu Kim, Penny Mealy, Bolor Erdene Munkhbayar, Claudia Ruiz Ortega, Ruozhi Song, and Lucas Zavala. Ansh Motiani served as a research assistant between January and March 2025. David Alzate, Conor Boyle, Eileen Burke, Ana Campos Garcia, Maria Arnal Canudo, Eliana Carranza, Claire Chase, Marie-Helene Cloutier, Jorge Coarasa, Nisan Gorgulu, Sabine Hertveldt, Margaux Lucrece Lelong, Xiaoyan Liang, Diego Luna Bazaldua, Daniel Mahler, Ana Maria Munoz Boudet, Klaus Oppermann, Ryan Milan Rafaty, Jamele Rigolini, Keiko Saito, Keiko Sakoda, Shane Sela, Dorothe Singer, Chandra Shekhar Sinha, Melvin Spreij, and Michael Weber are also contributors to the Report. Sandi Soe Lwin is providing administrative support.

The Report is sponsored by the Development Economics Vice-Presidency. Overall guidance for the preparation of the Report is provided by Indermit Gill, Senior Vice President and Chief Economist, and Somik Lall, DEC Interim Director for Development Policy.

Appendix B. Building on Past WDRs

This is the first *World Development Report* (WDR) focused on "standards." However, several previous WDRs have addressed standards and their impact on the development topics discussed.

[*Poverty and Human Development*](#) (1980), [*Investing in Health*](#) (1993), [*Knowledge for Development*](#) (1998/1999), and [*Learning to Realize Education's Promise*](#) (2018) touch upon standards in health care and education, focusing on areas such as standardization of drugs and supplies, quality assurance, accreditation, and certification.

[*Financial Systems and Development*](#) (1989), [*From Plan to Market*](#) (1996), [*Knowledge for Development*](#) (1998/1999), [*Entering the 21st Century*](#) (1999/2000), [*Attacking Poverty*](#) (2000/2001), [*Building Institutions for Markets*](#) (2002), and [*Finance for an Equitable Recovery*](#) (2022) examine various aspects of standards in financial institutions, including accounting practices, auditing procedures, banking regulations, and data dissemination.

[*The Challenge of Development*](#) (1991), [*Knowledge for Development*](#) (1998/1999), [*Entering the 21st Century*](#) (1999/2000), [*Building Institutions for Markets*](#) (2002), [*Agriculture for Development*](#) (2008), and [*Trading for Development in the Age of Global Value Chains*](#) (2020) focus on the role of standards in promoting trade, ensuring quality, and facilitating integration into global value chains. These Reports cover topics like international standards, open standards, sanitary and phytosanitary measures, and private standards.

[*Development and the Environment*](#) (1992), [*Investing in Health*](#) (1993), [*Entering the 21st Century*](#) (1999/2000), and [*Sustainable Development in a Dynamic World*](#) (2003) discuss standards in the context of environmental protection, covering topics such as setting and enforcing standards, incentivizing socially responsible behavior, and controlling pollution.

[*The Challenge of Development*](#) (1991), [*Workers in an Integrating World*](#) (1995), [*Attacking Poverty*](#) (2000/2001), and [*Jobs*](#) (2013) address labor standards, workplace safety, and the role of mandatory standards in protecting workers' rights. These Reports cover minimum wage laws, job security, core labor standards, and the enforcement of labor regulations.

[*Digital Dividends*](#) (2016) and [*Data for Better Lives*](#) (2021) discuss the importance of standards in ensuring data quality, promoting interoperability, and protecting privacy. These Reports cover data exchange standards, open standards, data protection standards, and the need for harmonization.

[*Poverty*](#) (1990) emphasizes the need for standardized approaches to measuring poverty. [*Sustainable Development in a Dynamic World*](#) (2003) and [*Making Services Work for Poor People*](#) (2004) emphasize the use of standards and benchmarks in competition law and service delivery reform. [*A Better Investment Climate for Everyone*](#) (2005) and [*Equity and Development*](#) (2006) focus on improving the investment climate and benchmarking country performance. [*Conflict, Security, and Development*](#) (2011) focuses on the role of standards in addressing leadership in fragile states. [*Risk and Opportunity*](#) (2014) discusses building codes and regulations to mitigate the impacts of disasters. [*Migrants, Refugees, and Societies*](#) (2023) highlights the use of standards to protect refugees internationally.

Appendix C. Standard-Setting Bodies

This appendix describes some of the main standard-setting bodies at the international, regional, and national levels. The year the body was established is indicated in parentheses following the name of the organization.

Treaty-based intergovernmental international organizations

- International Telecommunication Union (ITU) (1865): Focuses on information and communication technologies; membership includes public and private sectors; access to standards is free, but standards under development are available only to members.
- Union Postal Union (UPU) (1874): Sets technical standards for international mail delivery; membership includes public and private sectors; access to standards is free.
- International Labour Organization (ILO) (1919): Develops international labor standards; membership includes public and private sectors; access to standards is free.
- World Organisation for Animal Health (WOAH) (1924): Focuses on transparently disseminating information on animal diseases and improving animal health globally; WOAH is the WTO reference organisation for standards relating to animal health and zoonoses; access to standards is free.
- International Civil Aviation Organization (ICAO) (1944): Formulates standards and recommended practices for international aviation; membership includes public (member states) and private sectors; most standards are free, but some standards must be purchased.
- International Plant Protection Convention (IPPC) (1951): Develops, adopts and promotes the application of International Phytosanitary Measures (ISPMs) as the main tool to safeguard global food security, facilitate safe trade, and protect the environment; access to standards is free.
- International Organization for Legal Metrology (OIML) (1955): Focuses on legal metrology; membership includes public (member states) and corresponding members; access to standards is free.
- Codex Alimentarius Commission (1963): Develops international food standards, including sanitary and phytosanitary standards and guidelines; open to all members and associate members of the Food and Agriculture Organization (FAO) and World Health Organization (WHO); access to standards is free.

Nongovernmental international organizations

- International Electrotechnical Commission (IEC) (1906): Focuses on electrical, electronic, and related technologies; membership categories include Full, Associate, and Affiliate members, with Affiliates (developing countries) joining for free; access to standards must be purchased.
- International Organization for Standardization (ISO) (1947): Covers agriculture, construction, engineering, manufacturing, distribution, transport, health care, information technology (IT), environment, energy, safety, quality management, services, and so on; membership categories include Full, Correspondence, and Subscriber; membership fee is required to join ISO; access to standards require payment.

Neither treaty-based nor nongovernmental

- Basel Committee on Banking Supervision (1974): Focuses on banking standards; membership includes central banks and bank supervisors.

Private companies, industry and professional associations, and consortium standard-setting organizations

There are hundreds of standard-setting organizations. A few examples follow:

- American Society of Mechanical Engineers (ASME) (1880): Focuses on mechanical engineering areas; membership includes individuals and corporations; some standards are free but most require payment.
- Institute of Electrical and Electronic Engineers (IEEE) (1963): Covers energy, telecommunication, IT, electronics, robotics, automotive, health care, artificial intelligence (AI), and so on; membership includes public and private sectors; some standards are free but most require payment.
- World Wide Web Consortium (W3C) (1994): Develops international web standards; membership includes companies, institutions, and government entities; access to standards is free.
- Global Food Safety Initiative (GFSI) (2000): Establishes food safety management system standards; membership includes the food industry, government, academia, and service providers to the food industry; access to standards requires payment.
- ISEAL Alliance (2002): Focuses on social and environmental standards; serves as a sustainability standards system or accreditation body; access to standards requires purchase.

Regional standard-setting bodies

Many regions worldwide have regional standard-setting organizations, including the ASEAN Consultative Committee for Standards and Quality (ACCSQ) for the Association of Southeast Asian Nations (ASEAN); the African Regional Organization for Standardization (ARSO); the European Committee for Standardization (CEN); the European Committee for Electrotechnical Standardization (CENELEC); the Pan-American Standards Commission (COPANT); the Euro-Asian Council for Standardization, Metrology, and Certification (EASC); the European Telecommunications Standards Institute (ETSI); and the South Asian Regional Standards Organization (SARSO). Membership typically consists of National Standards Bodies (NSBs) in the region, and access to standards usually requires payment. These regional bodies develop standards across almost every social and economic sector, with standard-setting based on consensus among their member NSBs. Many standards are adopted or adapted from international standards.

National standard-setting bodies

Most countries in the world have recognized National Standards Bodies (NSBs), which are responsible for developing, coordinating, harmonizing, adopting, and disseminating standards at the national level. NSBs can be government organizations or private nonprofit organizations. Examples of government organizations include the Brazilian National Standards Organization (ABNT); the Bureau of Indian Standards (BIS); and the Bangladesh Standards Testing Institution (BSTI). Examples of private nonprofit organizations include the American National Standards Institute (ANSI) and the German Institute for Standardization (DIN). Membership typically includes industry associations, government agencies, scientific and research institutions, consumer groups, and professional societies. Access to standards usually requires payment. NSBs develop new standards and coordinate those created by their members, which can then be adopted as national standards. They also represent their countries at international standards forums, such as the International Electrotechnical Commission (IEC) and the International Organization for Standardization (ISO), adapting and incorporating many international standards developed by international standard-setting organizations into national standards.

Appendix D. External Consultations

African Organisation for Standardisation (ARSO), Beles Agri Food Lab (Ethiopia), Bureau of Indian Standards, Bureau of Philippine Standards (Department of Trade and Industry), Bureau Veritas, Central Bank of China, China Agriculture University, China Auto Information Technology Co., DQS Management Services, Edget Garment PLC (Ethiopia), Environmental Protection Authority of Ethiopia, Ethiopia Coffee and Tea Authority, Ethiopian Beverage Manufacturing Industries Association, Ethiopian Conformity Assessment Enterprise, Ethiopian Metrology Institute, Finternet, Gansu Longcuitang Nutrition and Health Food Co. (China), Gates Foundation, General Administration of Customs (China), Haimar Technology (Group) Co. (China), Hangzhou Juxing Technology Co. (China), Institute of Ethiopian Standards, International Electrotechnical Commission (IEC), International Labour Organization (ILO) (Country Office for China and Mongolia), International Organization for Standardization (ISO), International Telecommunication Union (ITU), Internet Corporation for Assigned Names and Numbers (ICANN), Korea International Cooperation Agency (KOICA) in Ethiopia, LaMP (Labor Mobility Partnerships), Malawi Bureau of Standards, Metrology University (China), Ministry of Business, Innovation, and Employment (New Zealand), Ministry of Environment and Ecology (China), Ministry of Foreign Affairs and Trade (New Zealand), Ministry of Trade and Regional Integration (Ethiopia), National Institute of Standardization (China), National Institute of Standards and Technology (NIST) (United States), Newland Food (China), Organisation for Economic Co-operation and Development (OECD), Pacific Islands Standard Committee (PISC), Peking University, Philippine Accreditation Bureau (Department of Trade and Industry), Rongzhi Social Responsibility Research Institute (China), Swiss State Secretariat for Economic Affairs, Theday Agro Industry (Ethiopia), United Nations Economic Commission for Europe, United Nations Industrial Development Organization, World Health Organization, World Intellectual Property Organization, and World Trade Organization.

Appendix E. Academic Advisory Committee and Other Academic Consultations

The Academic Advisory Committee is comprised of the following members: Emmanuelle Auriol (University of Toulouse), Henk de Vries (Erasmus University Rotterdam), Zonglai Kou (Fudan University), Adair Morse (University of California, Berkeley), Tim Simcoe (Boston University), Eric Verhoogen (Columbia University), and Cheryl Xiaoning Long (Xiamen University).

In addition, the WDR 2025 team has consulted with the following academics: Justus Baron (Northwestern University), Vittorio Bassi (University of Southern California), Solveig Bjørkholt (University of Oslo), Tim Büthe (Technical University of Munich), Jishnu Das (Georgetown University), Benedict Kingsbury (New York University), Giovanni Maggi (Yale University), Monika Mrazova (University of Geneva), Rebecca Perlman (University of California, Berkeley), Tommaso Porzio (Columbia University), Raghuram Rajan (University of Chicago's Booth School of Business), Camille Reverdy (Geneva Graduate Institute), Shaoda Wang (University of Chicago), and Paul Wiegmann (Eindhoven University of Technology).

Appendix F. High-Level Advisory Panel

Andrea Maechler (Deputy General Manager, Bank for International Settlements [BIS]), Member to be identified from World Trade Organization [WTO]),* Sergio Mujica (Secretary-General, International Organization for Standardization [ISO]), Nandan Nilekani (Co-Founder and Chairman, Infosys Technologies Limited), Monika Weber-Fahr (Chair, Board of Directors, ISEAL Alliance), Adil Zainulbhai (Chairman, Capacity Building Commission of India), and Yongmei Zhou (Professor, National School of Development & Institute of South-South Cooperation and Development, Peking University).

Appendix G. Background Papers

History of Standards
China's Evolution in International Standardization: From Follower to Global Leader
From Standards Adopter to Standards Author: The Case of South Korea
The Political Economy of Safety Standards: France and Britain in the 19th Century
Standard Sizes: The Government's Role in Diffusing Mass Production Techniques in the U.S., 1917-1960s
Conceptual Framework
Quality Infrastructure Maturity Trajectories: Experiences and Prospects of Advanced and Developing Economies
Evolution of India's National Quality Infrastructure
Using Standards to Promote Sustainable Development
Standards for a Better Economy
Digital Public Infrastructure (DPI) in India: Setting Standards with the Hourglass Model
The Development of Agrifood Standards
Ethiopia's Experience with Internal Standards vs. International Standards
Standards for a Better Society
Setting Norms and Standards in Global Health: Challenges and Opportunities
Standards for a Better Environment
Extreme Heat and Cooling Standards for Climate Change Adaptation
Standards and the Regulation of Plastic Wastes in Developing Countries

Notes

¹ According to the United Nations, least developed countries are low-income countries facing severe structural challenges that hamper their development. For more information, please see <https://www.un.org/development/desa/dpad/least-developed-country-category.html>.

² Refer to Rauch and Evans (2000) and World Bank Group (2019) for HRM; and Hanson et al. (2022) and Upadhaya et al. (2024) for PFM.

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